Particulars of the Experiments Performed

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2	8/12/2025	Measurements of Screw thread parameters using tool maker's microscope (Shadow Image method)	13 - 17	20		
3	27/10/25	Measurement of straightness using Autocollimator/Laser interferometry. Gage R & R using MiniTab.	18 - 23	20		
4		Measurement of Gear Tooth Thickness Using Gear Tooth Vernier Caliper.	24 - 28	20		
5		Measurement of roughness parameters using Dial indicator and Surftester.	29 - 34	20		
6	13/10/25	Calibration of Thermocouple. Gage R & R using MiniTab.	35 - 39	20		
7	05/01/26	Calibration of Linear Variable Differential Transformer (LVDT)	40 - 46	20		
8		To obtain the characteristics of LVDT inductive transducer and determine its sensitivity and Linearity	47 - 49	15		
9		Programming and Simulation of Bottle-filling process using PLC.	50 - 52	15		
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11		Project Based Learning	56 - 58	20		
Total				200		
TEST				50		
Total				250		
Final Lab Marks				50		

Effective diameter = T + K

Where,

T= Dimension Under the wire and,

$$T = D \pm (R_s \sim R)$$

Where

D = Diameter of setting cylinders

R_s = reading over standard with wire

R = reading over specimen with wire (Average V

K is the value which depends upon diameter of the wire (d), pitc thread (P) and angle of the screw thread (θ)

$$K = \frac{P}{2} x \cot(\frac{\theta}{2}) - d\,(\text{cosec}(\frac{\theta}{2}) - 1)$$

Here, $\theta = 60^0$ for metric thread

For Metric Thread, K = 0.866p – d

Where,

P = pitch of the screw thread, d = diameter of wires which depend the screw thread

Least Count Calculations:

$$\frac{\text{Pitch}}{\text{No. of divisions}} = \frac{0.5}{250} = 0.0002\,mm$$

Vernier Scale LC = 9/10 of thimble

$= 9/10 \times 0.002$

LC of instrument = IMSD – IVSD

$= 0.002 - 0.0018$

$= 0.0002\,mm$

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Observations:

1. Pitch of the given screw thread (P) = 2mm
2. Thread angle of the given screw thread = 60°
3. Wire size used (d) = 1.35mm
4. Setting cylinder diameter (D) = 8mm
5. Least count of instrument = 0.0002mm

Tabulation:

1. **Major Diameter**

Sl. No	Position	R_s (mm)	R (mm)	$R_{Average}$	$(R_s \sim R)$
1	I	3.5228	14.0194	14.0168	10.494
	II	3.5228	14.0168		
	III	3.5228	14.0144		

2. **Minor Diameter**

Sl. No	Position	R_s (mm)	R (mm)	$R_{Average}$	$(R_s \sim R)$
2	I	9.5482	18.0326	18.0335	8.4853
	II	9.5482	18.0372		
	III	9.5482	18.0308		

3. **Effective Diameter**

Sl. No	Position	R_s (mm)	R (mm)	$R_{Average}$	$(R_s \sim R)$
3	I	6.0376	15.0484	15.0476	9.0100
	II	6.0376	15.0478		
	III	6.0376	15.0466		

Signature of staff in-charge

Specimen Calculation:

Effective diameter = T + K

Where,

T= Dimension Under the wire and,

$$T = D \pm (R_s \sim R)$$

Where

D = Diameter of setting cylinders = 8mm

R_s = reading over standard with wire=

R = reading over specimen with wire (Average Value)=

T = 17.01mm

=

T = 17.01mm

and,

K is the value which depends upon diameter of the wire (d), pitch of the sc thread (P) and angle of the screw thread (θ)

$$K = \frac{P}{2} x \cot(\frac{\theta}{2}) - d\,(\text{cosec}(\frac{\theta}{2}) - 1)$$

Here, $\theta = 60^0$ for metric thread, p = 2mm , d = 1.35mm

For Metric Thread, K = 0.866p – d

Where,

P = 2mm , d = 1.35mm

K = 0.866p – d

= 0.866(2) ~1.35

K = 0.382mm

E = T + K

= 17.01 + 0.382

$T = D \pm (R_s - R)$

$= 8 \pm (9.0100)$

T = 17.01mm

Effective Diameter **E = T + K** = 17.392mm

Major diameter = **D ± (R_s ~ R)** = 8 ± (10.494) = 18.494mm

Minor Diameter = **D ± (R_s ~ R)** = 8 ± (8.4853) = 16.4853mm

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OBSERVATION / DATA SHEET

Date 03/11/2025 Name Sudhama. Sinha

Dept./Lab IEM Class IIIrd Sem Expt./No. E01

Title Floating carriage Applications & Quality Insights.

Scrum-1:

Major ϕ

Position	Rs (mm)	R(mm)	Rang	Rs - R
I	4.2808	14.8148		
II		14.8018	14.7995	10.5187
III		14.7820		

Minor ϕ

Position	Rs (mm)	R(mm)	Rang	Rs-R
I	10.5948	18.9460		
II		18.9300	18.9288	8.3340
III		18.9104		

Effective ϕ

Position	Rs (mm)	R(mm)	Rang	Rs-R
I	6.9336	16.0342		
II		16.0436	16.0484	9.1148
III		16.0376		

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Teacher Incharge

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OBSERVATION / DATA SHEET

Date 03/11/2025 Name Sudhanva. Sneha

Dept./Lab IEM Class 3rd sem Expt./No. E01

Title Floating carriage

rue 3:

lajor ϕ

Position	Rs(mm)	R(mm)	Rawg.	Rs-R
I	3.91	12.64	12.6366	8.7266
II		12.6348		
III		12.6350		

Minor ϕ

Position	Rs(mm)	R(mm)	Rawg	Rs-R
I	9.1200	15.9402	15.9348	6.8148
II		15.4310		
III		15.9332		

Effective ϕ

Position	Rs(mm)	R(mm)	Rawg	Rs-R
I	5.8400	13.7804	13.7884	7.9484
II		13.7922		
III		13.7926		

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Screw 1:

Major ϕ = 10.5187 mm

Minor ϕ = 8.3340 mm

Effective ϕ = 9.1148 mm

Screw 2:

Major ϕ = 13.3402 mm

Minor ϕ = 8.9346 mm

Effective ϕ = 10.5880 mm

Screw 3:

Major ϕ = 8.7266 mm

Minor ϕ = 6.8148 mm

Effective ϕ = 7.9884 mm

2) The measured major, minor and effective diameters compared with specified tolerance limits to determ acceptance or rejection. These values are further used to monitor process stability and capability. The e diameter, having the highest tolerance, acts as the primary for functional acceptance of threaded a

Results:

The effective diameter, major diameter and the minor diameter of the given screw thread specimen are determined using two wire method and are as follows:

(1) The Major of given screw thread is = 18.494mm

(2) The Minor diameter of given screw thread = 16.4853mm

(3) The Effective diameter of given screw thread is = 17.392mm

Inference: Floating carriage diameter measuring machine is designed to have accurate measurement of major, minor & effective diameter of precision. Screw threads as well as those of odd flutes screw taps, it works on principle of nut & bolt screw threads. It can be used to check effective diameter of taps.

Applications & Quality Insight:

1) Take three samples of a screw and measure threads, then calculate mean, standard deviation, and % deviation from nominal.
2) Discuss how this relates to tolerance, SPC, and acceptance/rejection criteria in Quality Assurance.

Particulars	Marks Awarded	Marks Obtained
1. Conceptual Understanding	04	
2. Conduction of Experiments	04	
3. Write up & Performance Accuracy	04	
4. Inference	04	
5. Viva	04	
Total	**20**	
Signature with Date		

Thread Angle:

1. Rotate the screen until a line on the screen coincides with one flank of profile and note the angle of the screen rotation(α_1')
2. Rotate the screen further till the same line coincides with the other flank of and note down the angle.(α_1'')
3. Take two similar angular readings on the other side of the thread profile ima α_2'')
4. Calculate the angle of the thread using suitable formulae.

Formulae Used:

- Actual Reading = $I_R \sim F_R$

  Where,

  I_R = Initial Reading

  F_R = Final Reading

- Thread Angle, $\alpha = (\alpha_1 + \alpha_2)$,

Where,

$$\alpha_1 = (\alpha_1' - \alpha_1'')/2$$

$$\alpha_2 = (\alpha_2' - \alpha_2'')/2$$

Least Count Calculation:

Pitch of micrometer screw = 0.2mm

No. of division on circular scale = 200

$$\text{Least count (LC)} = \frac{\text{Pitch}}{\text{No. of scale division}}$$

$$= \frac{0.2}{200}$$

$$LC = 0.001\text{mm}$$

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OBSERVATION / DATA SHEET

Date 8/12/2025 Name Sudhanva-Smitha

Dept./Lab DM Class IEM Expt./No. E02

Title TMM

Screw: 1

Parameter	Initial I_R(mm)	Final F_R(mm)	Actual $I_R - F_R$
Major ϕ	1.720	23.520	21.800
Minor ϕ	3.400	21.900	18.500
Effective ϕ	1.720	21.900	20.180
Pitch	10.900	13.400	2.500
Depth	1.720	3.350	1.630
Angle of thread	$\alpha_1 = 359°50'$	$\alpha_2 = 29°50'$	$\alpha_3 = 60°00'$

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Screw: 2

Parameter	Initial IR(mm)	Final FR(mm)	Actual (IR-FR)
Major ϕ	2.150	26.450	24.300
Minor ϕ	4.200	24.300	20.100
Effective ϕ	2.150	24.300	22.150
Pitch	12.600	15.600	3.00
Depth	2.150	4.00	1.850
Angle of thread	$\alpha_1 = 358°40'$	$\alpha_2 = 298°40'$	$\alpha = 60°00'$

Screw: 3

Parameter	Initial IR(mm)	Final FR(mm)	Actual (IR-FR)
Major ϕ	1.500	20.700	19.200
Minor ϕ	3.00	18.900	15.900
Effective ϕ	1.500	18.900	17.400
Pitch	9.000	11.00	2.00
Depth	1.500	2.900	1.400
Angle of thread	$\alpha_1 = 359° 30'$	$\alpha_2 = 299° 30'$	$\alpha = 60°00'$

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OBSERVATION / DATA SHEET

Date 8/12/2025 Name Sudharsana Senthil

Dept./Lab DM Class IEM Expt./No. E02

Title TMM

Mean of the parameters:-

major ϕ = (21.800+24.300+19.200)/3 = 21.766

minor ϕ = (18.500+20.100+15.400/3 = 18.166

Equitie ϕ = (20.180+22.150+17.400)/3 = 19.91

Pitch = (2.500+3.000+2.000)/3 = 2.5

Depth = (1.630+1.850+1.400)/3 = 1.626

Angle of thread = (60+60+60)/3 = 60.00

Standard deviation of parameters:

major ϕ = μ = 21.766

$(x-\mu)$ = 0.034, 2.534, -2.566

$(x-\mu)^2$ = 0.00156, 6.421, 6.584

$S^2 = \frac{13.006}{3} = 4.335$

$S = \sqrt{4.335} = 2.082$

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Teacher Incharge

Minor $\phi = \mu = 18.166$

$(x-\mu)^2 = 0.1115, 3.740, 5.134$

$s^2 = 2.995$

$s = 1.73$

Effective $\phi = \mu = 19.91$

$(x-\mu)^2 = 0.00729, 5.0176, 6.3$

$s^2 = 3.7968$

$s = 1.948$

Pitch $\Rightarrow \mu = 2.5$

$(x-\mu)^2 = 0, 0.25, 0.25$

$s^2 = 0.1666$

$s = 0.4081$

Depth $\Rightarrow \mu = 1.626$

$(x-\mu)^2 = 0.000016, 0.0501, 0.051$

$s^2 = 0.337$

$s = 0.1835$

Angle of thread $\Rightarrow \mu = 60^\circ$

$(x-\mu)^2 = 0, 0, 0$

$s^2 = 0$

$s = 0$

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OBSERVATION / DATA SHEET

Date 8/12/2025 Name Sudhanva Savitha

Dept./Lab DM Class IEM Expt./No. E02

Title TMM

* In quality assurance (QA), the measurements obtained from a toolmaker's microscope are used to decide whether a threaded component meets design specification and can be accepted.

A threaded component is accepted only if:

* All measured parameters lie within specified tolerance limits.
* Thread profile matches standard (ISO/metric)
* No functional interference during assembly.

* TMM provides high resolution repeatable measurement of thread geometry.

Signature of
Teacher Incharge

By tracking dimensional changes over time and through can be quantified and is:-

* Predict remaining service life.
* Schedule condition based maintainence.
* Prevent sudden failure.

By analysing dimensional trends and wear rate, remaining life. Can be predicted and maintained proactively ensuring reliability, safety and cost effective operation.

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Observations:

Least count of Tool maker's microscope = 0.001 mm

Tabulation:

Sl. No.	Parameter	Initial reading I_R (mm)	Final reading F_R (mm)	Actual reading = I_R~F_R (mm)
1	Major diameter	1.792	23.595	21.803
2	Minor diameter	3.426	21.986	18.56
3	Effective diameter	1.792	21.986	20.194
4	Pitch	10.941	13.455	2.514
5	Depth of thread	1.792	3.426	1.634
6	Angle of screw thread	$\alpha_1' = 359^\circ - 60'$ $\alpha_1'' = 331^\circ - 25^\circ$	$\alpha_2' = 30^\circ - 19$ $\alpha_2'' = 29^\circ - 41$	$\alpha = 60^\circ 00'$
		29.41	59.6 ≃ 60°	

Signature of staff in-charge

Results:

Sl. No.	Screw thread Parameter
1	Major diameter = 21.803
2	Minor diameter = 18.56
3	Effective diameter = 20.194
4	Pitch = 2.514
5	Depth of thread = 1.634
6	Angle of screw thread = 60°00'

Inference:
(comment on variation between readings and reason for any discrepancies)
minor variation is observed in some thread measurements. This is mainly due to alignment and focusing issues in the shadow image method. Instrument limitations

Thread Surface Quality and Defect Assessment Using TMM
Task:

1. Take three used or worn screws and examine the thread profile using TMM.
2. Measure major diameter, minor diameter, and thread depth, noting any irregular wear, or surface defects.
3. Calculate the mean, standard deviation, and % deviation from nominal values.

Quality Insight:

- Identify whether deviations are due to manufacturing errors or wear.
- Discuss how these measurements could guide acceptance/rejection of component quality assurance.
- Suggest how TMM data can be used to predict remaining service life or schedu preventive maintenance.

Particulars	Marks Awarded	Marks Obtained
1. Conceptual Understanding	04	
2. Conduction of Experiments	04	
3. Write up & Performance Accuracy	04	
4. Inference	04	
5. Viva	04	
Total	**20**	
Signature with Date		

Procedure:

1. Divide the surface plate into finite number of bridge lengths and each bridge len
   to 100 mm.
2. Adjust the Auto collimator screws to coincide the Graticule with Eyepiece Gratic
3. Note down the corresponding reading from horizontal micrometer i.e. as a initia
   (say 10,0)
4. Place the Reflector on 1st position such that length equal to 1st bridge length (l
   and observe the light beams. If light target and eyepiece graticule are not coincid
   adjust the horizontal micrometer to coincide both Graticule.
5. Note down the corresponding reading from horizontal micrometer i.e. as final readi
6. Calculate the difference of readings to obtain angular deviations.
7. Calculate the Linear deviation using the suitable formula.
8. Repeat the same procedure for different positions.
9. Plot a graph of position of reflector v/s deviation

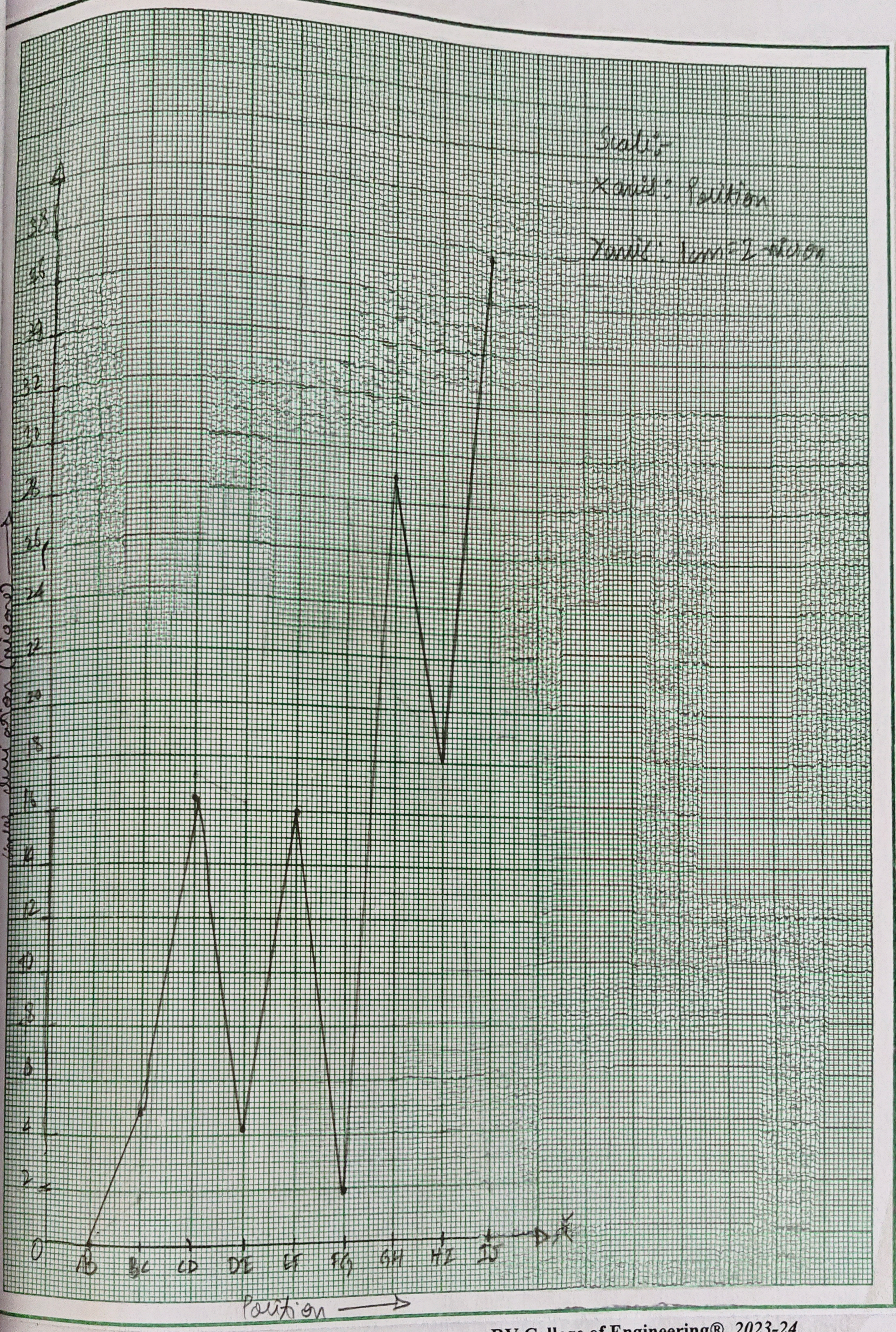
Formula Used:

Linear deviation = tan (angular deviations in degree) * 100 mm
= Angular deviations in rad * 100 * 1000 μ

Least Count Calculation:

Least count of autocollimator = $\frac{2\,min}{120\,sec} \rightarrow \frac{120\,sec}{120\,sec}$

= 1 second


Static
Condition
Condition: Item = 2 units
Displacement (units)
Position
0
AB
BC
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